An Instrumental Variable Approach to Dynamic Models

work in progress, comments welcome

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Introduction

In this paper, we propose an instrumental variable approach to the estimation of dynamic models, both single and multiple agent. The IV approach allows us to deal with endogenous market structure in an economically meaningful way. The approach can (in principle) handle models that are incomplete and/or set identified because of some combination of initial conditions, multiple equilibria and discrete data.

The approach has close connections to "two-step" methods that build on Hotz-Miller as well as to more computationally based methods that lead to MLE estimation.

I.I.D. Shocks

In IO, much of the existing empirical literature models unobserved shocks as private information, independently distributed over time.

This greatly simplifies the estimation methods, as outlined in a series of influential papers including Rust (1987), Hotz and Miller (1993) Bajari, Benkard and Levin (2007) , Pakes, Ostrovsky, and Berry (2007) and Pesendorfer and Schmidt-Dengler (2008).

Independent Private Shocks in Dynamic Models

Given an independent private information assumption, the current state does not reflect any (persistently) unobserved factors, only accumulated past luck. So, the current state is econometrically exogenous.

This leads naturally to the two-step Hotz-Miller style approaches:

1. first identify the dynamic policy function, "directly from the data" and then
2. use this plus Bellman's equation to identify structural parameters of the single-period return function.

For formal identification, see Magnac and Thesmar (2002) and related literature.

Problems with Exogenous States

But the treatment of states as econometrically exogenous may be problematic. It is greatly at odds with the focus, in associated static literatures, on the econometric endogeneity of market structure.

Think of possible states in an IO model: past entry, number of firms or outlets, capital, quality .... These are associated with various dynamic decisions: entry, store opening, investment in capital or quality. In each case, it seems likely that persistent unobservables are correlated with both the current decision and the current observed state.

This introduces a classic endogeneity issue into the identification/estimation of the dynamic policy function. Classic solutions for endogeneity include panel data and IV methods.

Serial Correlation in Optimizing Dynamic Models

- Keane and Wolpin (1997) (single agent discrete-type, no strict identification proofs, moderately computational intensive)
- Arcidiacono and Miller (2011) (oligopoly with mixture models and possible multiple equilibria, computationally intensive, no strict identification proofs, but see Kasahara and Shimotsu (2009).)
- Pakes and Ericson (1996) (oligopoly; similar to us, but "fit to ergodic distribution" for us becomes "fit to IV restrictions").
- Kalouptsidi, Scott, and Souza-Rodrigues (2018) (single-agent special case, same intuition for IV with simple linear implementation)

Literatures on "less structural" dynamic panel models include important work on persistent heterogeneity vs. state dependence and on initial conditions problems

An IV Alternative

At the most basic level, we adopt the classic two-step approach but identify (or at least restrict) the policy function via generalized instrumental variable ("GIV") methods following the exposition in Chesher and Rosen (2017).

Chesher and Rosen (2017) builds on a very large prior and on-going literature, including many early papers on incomplete models by Manski (*e.g.* 2003) and/or Tamer, work on discrete entry models by Tamer and also work on sharply identified sets such as by Beresteanu, Molinari, and Molchanov (2011) & Galichon and Henry (2011).

Chesher and Rosen (2017) is useful to us both because of the generality of the approach but also largely for the exposition that focuses on IV intuition in a broad class of models with discrete outcomes and possibly incomplete models.

IV Intuition

The IV intuition is that past exogenous variation will be correlated with current states. E.g. if Detroit was large and rich 50 years ago, it may have many Sears stores today. Past macro shocks may effect today's market structure, and these may interact with market-level characteristics. Past regulatory regimes may be correlated with market structure (consider hospitals).

In general, natural IVs include *past exogenous cost and demand shifters* that (in the presence of sunk costs, etc) have led to the present endogenous state.

Outline

Today

- Dynamic Model (notation & examples)
- GIV restrictions
- Identified Set of Policies
- Example: Single Firm Entry
  - Identified sets as serial correlation, IV strength & T vary
  - Inference example
- Oligopoly
  - Broad Idea
  - Worked example with both serially correlated public and i.i.d private info (skip today)
- Only the beginning of a data example, using Collard-Wexler data

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Model

We begin with a single firm model. We see a large set of markets, each with a single firm, for a small fixed T.

Then we consider the oligopoly case.

Single Period Return

Monopoly Model

Panel data for a (fixed) T periods, with the periods denoted $t = 1, \ldots, T$. The full set of variables are not necessarily available for any prior history of the firm, $t < 1$, although there may be some partial history.

Single Period Profits for market i: $\pi\left(a_{it}, x_{it}, w_{it}, u_{it}; \theta_{\pi}\right)$

- x_{it} endogenously chosen state(s)
- a_{it} choice (policy) variable(s) (action)
- w_{it} exogenous profit shifter(s)
- $u_{it} \in \mathbb{R}$, unobserved (to us) serially correlated unobservable

Any additional exogenous variables correlated with x_{it} (i.e. policy prior to $t = 1$) are denoted r_i. Could extend (at cost) to multiple unobservables (esp. with multiple actions).

Further notation

$$
\begin{aligned}
x_i &= (x_{i1}, \ldots, x_{iT}) \in \mathbb{X} \\
a_i &= (a_{i1}, \ldots, a_{iT}) \in \mathbb{A} \\
w_i &= (w_{i1}, \ldots, w_{iT}) \in \mathbb{W} \\
u_i &= (u_{i1}, \ldots, u_{iT}) \in \mathbb{U}
\end{aligned}
$$

Transitions

Classic setup with added serial correlation

Endogenous States:

$$\Gamma(x_{it+1}|a_{it}, x_{it})$$

Could be deterministic (degenerate) or stochastic state transitions.

Exogenous Observed States

$$\Lambda(w_{it+1}|w_{it})$$

Unobserved (by us) States

$$\Phi(u_{it+1}|u_{it}; \theta_u)$$

Assume Γ, Λ are known and/or identified from data, but θ_u is unknown.

Examples

Table: Some Single Agent IO Examples

State, x_{it}	Action, a_{it}	$\mathcal{A}(x_{it})$	Transition
Capital	Investment	$\mathbb{R}^+$	$x_{it+1} = \lambda x_{it} + a_{it}$
Out/In	Entry/Exit	$\{0, 1\}$	$x_{it+1} = a_{it}$
Retail	# of Stores	$\mathcal{I}^+$	$x_{it+1} = a_{it}$
Quality	R&D	$\mathbb{R}^+$	$x_{it+1} \sim f\left(x_{it}, a_{it}\right)$

IV Assumption

$$z_i = (r_i, w_{i1}, w_{i2}, \ldots, w_{iT}).$$

$$u_i = (u_{i1}, u_{i2}, \ldots, u_{iT}).$$

Independence of the instrument and the unobservables:

$$z_i \perp u_i.$$

Possible Instruments?

State	Example IVs?
Capital	Past investment cost, past productivity shock
Out/In of Market	past market population, past zoning
# of Stores	distance from headquarters, interacted with time?
Quality	Past R&D shocks, age of firm

Dynamic Problem

Bellman Equation:

$$V\left(x_{it}, w_{it}, u_{it}\right) =$$

$$\max_{a_{it} \in \mathcal{A}(x_{it})} \left(\pi\left(a_{it}, x_{it}, w_{it}, u_{it}, \theta_{\pi}\right) + \delta E\left[V\left(x_{it+1}, w_{it+1}, u_{it+1}\right) | a_{it}, x_{it}, w_{it}, u_{it}\right]\right.$$

where

$$E\left[V\left(x', w', u'\right) | a, x, w, u\right] =$$

$$\int\int\int V\left(x', w', u'\right) d\Gamma(x'|a, x) d\Lambda(w'|w) d\Phi(u'|u; \theta_u).$$

Policy function

In the true model (and therefore the data), the policy function takes the form

$$a_{it} = \sigma\left(x_{it}, w_{it}, u_{it}\right) \in \mathcal{F}.$$

There may be qualitative restrictions (monotonicity) embodied in the set of functions $\mathcal{F}$.

For a particular $\theta = (\theta_\pi, \theta_u)$, the policy function derived from known transitions and Bellman's equation is:

$$a_{it} = \sigma_\theta\left(x_{it}, w_{it}, u_{it}\right).$$

Identification

We focus first on identification.

For purposes of identification, we assume that we observe the true data generating process, across firms or agents, denoted

$$P(a_i, x_i, w_i, r_i).$$

This is equivalent to seeing T period panel on a very large (in fact, infinite) cross-section of firms or agents.

The unknowns are the parameters of profits θ_π and θ_u. Nothing in our general discussion of identification requires these to be finite dimensional, but in practice we consider only finite-dimensional parametric models (including fully flexible profit functions with discrete (a_i, x_i, w_i)).

Incomplete Model

$$
\begin{aligned}
a_{iT} &= \sigma\left(x_{iT}, w_{iT}, u_{iT}\right) \\
x_{it} &\sim \Gamma\left(x_{iT-1}, a_{iT-1}\right), \\
a_{T-1} &= \sigma\left(x_{iT-1}, w_{iT-1}, u_{T-1}\right) \\
x_{T-1} &\sim \Gamma\left(x_{iT-2}, a_{T-2}\right), \\
&\vdots \\
a_1 &= \sigma\left(x_{i1}, w_{i1}, u_{i1}\right).
\end{aligned}
$$

But: there is no model of the endogenous x_{i1}, which is inherited from unobserved prior history. If period 1 was the "birth" of the agent, we might have a model for x_{i1}, which would complete the single agent model. (Of course, this still doesn't guarantee identification.) Oligopoly will introduce a second source of incompleteness: multiple equilibria.

Chesher and Rosen GIV

Idea: (set) identify the policy function from classic instrumental variables conditions, extended to "Generalized IV" (GIV) to deal with

1. incomplete model (we only have necessary or sufficient conditions on the unobservables, not necessary and sufficient) and
2. discrete variables, as in entry/exit models
3. lack of point identification of the parameters, even in the absence of problems 1 and 2.

We may have all or none of these problems. Discrete actions naturally lead to conditions on sets of unobservables that give a particular policy a_{it}.

Necessary Conditions

a familiar step

If the sequence (x_i, w_i, a_i) occurs, then u_i must be in the inverse image set

$$\mathcal{U}(a_i, w_i, x_i, \sigma) = \{u_i : \sigma(x_{it}, w_{it}, u_{it}) = a_{it}, \forall t\}$$

The condition $\{u_i \in \mathcal{U}(a_i, w_i, x_i, \sigma)\}$ is then a necessary condition for event (x_i, a_i).

Policy Functions consistent with IV conditions

Using Cheshire-Rosen style notation, a pair $(\sigma(x_{it}, w_{it}, u_{it}), \theta_u)$ is then in the identified set iff for all closed sets $\mathcal{S} \in \mathbb{U}$ and $\forall z$

$$\Pr\big(\mathcal{U}(a_i, x_i, w_i, \sigma) \subseteq \mathcal{S} \mid z\big) \leq \Phi(\mathcal{S}; \theta_u) \tag{1}$$

The LHS is the conditional probability of the outcomes $y_i = (a_i, x_i)$ that, according to σ, have $\{u_i : u_i \in \mathcal{S}\}$ as a necessary condition. For a given σ and z, this probability is observed in the data. The RHS is the probability of that necessary condition wrt the distribution of u_i. The RHS is determined by the distribution of u_i, which by assumption does not depend on z.

But what "test sets" $\mathcal{S}$ to use?

Chesher and Rosen Sharply Identified Policy Functions

CR show that to obtain the sharply identified θ set we only need to check certain sets $\mathcal{S} \in Q(\sigma, w_i)$.

The collection of sets $Q(\sigma, w_i)$ is the "core determining set" as defined in CR (2017) and earlier work, the minimal collection of closed sets $\mathcal{S} \in \mathbb{U}$ that yields the sharp identified set for θ. These include the overlapping sets of $\mathcal{U}(a_i, x_i, w_i, \sigma)$, excluding cases of strict subsets. For simple low dimensional discrete problems, $Q(\sigma, z_i)$ can be easy to compute and not "too big", but it can otherwise grow very (indeed infinitely) large.

This result tells us the necessary conditions we need to check to get sharp identification. Perhaps most useful for smaller problems and to build intuition in examples.

Equalities and Complete Models

In some cases, some of the inequalities in (1) are equalities, because the necessary conditions are *necessary and sufficient* for particular (sets of) actions.

In a complete model, all of the conditions would be equalities. However, as usual, this does not guarantee that the parameters are point identified.

Policies identified by GIV Alone

For a given θ_u, denote by

$$\Sigma^{IV}(\theta_u) \subseteq \mathcal{F}.$$

the set of σ functions identified from the data and the IV restrictions—*i.e.* those that satisfy condition (1) $\forall \mathcal{S} \in Q(\sigma, z)$ and $\forall z$. (Note when $\Sigma(\theta_u) = \varnothing$, for some θ_u, then that value of θ_u is excluded from the identified set without any use of the Bellman equation.)

Note that the sets Σ^{IV} are determined exclusively by the IV conditions and the data, with no use of the dynamic model. Making use of Bellman's equation will further shrink the set of identified policies (see discussion below).

Identified Set of Structural Parameters

For any $\theta = (\theta_\pi, \theta_u)$, we can use the Bellman equation to compute the implied policy $\sigma_\theta(x_{it}, w_{it}, u_{it})$. For $\theta = (\theta_\pi, \theta_u)$, this policy is

$$\sigma_\theta(x_{it}, w_{it}, u_{it}) \equiv$$

$$\underset{a_{it} \in \mathcal{A}(x_{it})}{\operatorname{argmax}} \Big(\pi(a_{it}, x_{it}, w_{it}, u_{it}) + \delta E\big[V(x_{it+1}, w_{it+1}, u_{it+1}) | a_{it}, x_{it}, w_{it}, u_{it}\big] \Big).$$

The sharply identified set of parameters is then

$$\Theta_{ID} \equiv \{\theta = (\theta_\pi, \theta_u) \colon \sigma_\theta(x_{it}, w_{it}, u_{it}) \in \Sigma^{IV}(\theta_u)\} \tag{2}$$

This imposes both the dynamic model and the GIV restrictions. This is the sharply identified set because any θ in this set generates a policy function that cannot be rejected by the data plus the IV condition.

Connection to Two Step Models

The prior exposition suggests that we could

1. identify a (set of) policy function(s) that are consistent with the data and the IV restrictions and then
2. see which structural parameters are consistent with the identified policy function(s). These are the identified parameters.

These are the same steps as in the 2-step dynamic literature with independent shocks (following on Hotz-Miller). The complications here are the presence of parameters for the unobservables (necessary at least to model the degree of serial correlation) and the IV methods in the first step, as opposed to directly "fitting the policy to data." The methods are the same when the distribution of u_t is known (or normalized) and the states are exogenous (independent errors.)

Connection to Two Step Models

Continued

Under appropriate assumptions, one might point identify the policy function in step 1. In this case, our procedure is just like a 2-step model, except that one runs GIV first stage (vs assuming exogenous covariates.)

For example, when the choice variable is continuous, point identification of the policy is attained under the assumptions in Chernozhukov and Hansen (2005).

When the choice variables is discrete, often one only gets partial identification of the policy (Chesher (2010); Chesher, Rosen and Smolinski (2013)).

Connection to Two Step Models

Continued

However, as in the i.i.d. literature, there is no reason to proceed in 2 steps, in that order.

As a computational alternative, one could search over the space of θ's, for each possible θ

1. computing $\sigma_\theta(x_{it}, w_{it}, u_{it})$ via the contraction mapping and then
2. testing whether σ_θ survives the IV restrictions applied to the data. If so, that particular θ is in the identified set, otherwise not.

Example: Single Firm "Entry / Exit"

A minimal model to think about dynamics.

$$a_{it} \in (0, 1)$$

State is In/Out in the prior period

$$x_{it} \in (0, 1)$$

Single period payoffs, $\pi(a_{it}, x_{it})$, have a random shock, ϵ_{it}, plus a scalar sunk entry cost, γ.

$$\begin{aligned}
\pi(0,1) &\equiv \pi(0,1) = 0, \\
\pi(1,1) &= \bar{\pi} - \epsilon_{it}, \text{ with} \\
\pi(1,0) &= \pi(1,1) - \gamma, \text{ and}
\end{aligned}$$

$$\epsilon_{it} = \rho\epsilon_{i,t-1} + \nu_{it},$$

where $\nu_{it} \sim \mathcal{N}(0, 1)$. Think of u_{it} as the quantile of ϵ_{it}.

3 structural parameters: $\bar{\pi}, \gamma, \rho$.

Policy Function in the Example.

Policy function:

$$a_{it} = \sigma(x_{it}, u_{it})$$

where $u_{it} \sim \text{unif}(0, 1)$.

The dynamic model generates a monotonicity result that σ is weakly increasing in x and weakly decreasing in u.

Policies Identified by GIV from only One Transition

single firm entry/exit example

This is a binary choice style model *a la* Chesher (2010), using only the marginal distribution of u, which is independent of some instrument.

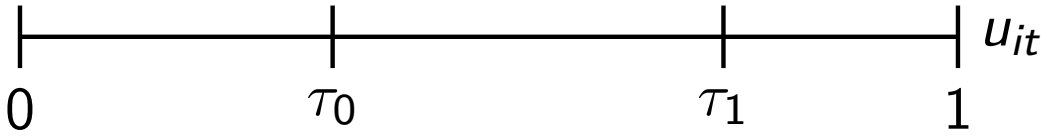


Figure: Policy Cut-offs in the Example

As an example of identification, we look at restrictions only involving the marginal distribution of u, next consider IV restrictions on the joint distribution across periods.

Restrictions Using Only One Choice

Table: Inverse image sets for the Binary-Binary Marginal Example

a	x	$\mathcal{U}(a,x)$
1	1	$(0,\tau_1)$
1	0	$(0,\tau_0)$
0	1	$(\tau_1,1)$
0	0	$(\tau_0,1)$

Table: Restrictions via Elemental Sets for the Binary-Binary Marginal Example

$\mathcal{S}$	$\Pr\big(\mathcal{U}(a_i,x_i,\sigma)\subseteq\mathcal{S}\ \vert z\big)$	$\leq\Phi(\mathcal{S};\theta_u)$
$\mathcal{U}(1,1)$	$\Pr((1,1)\vert z)+\Pr((1,0)\vert z)$	$\leq\tau_1$
$\mathcal{U}(1,0)$	$\Pr((1,0)\vert z)$	$\leq\tau_0$
$\mathcal{U}(0,1)$	$\Pr((0,1)\vert z)$	$\leq 1-\tau_1$
$\mathcal{U}(0,0)$	$\Pr((0,0)\vert z)+\Pr((0,1)\vert z)$	$\leq 1-\tau_0$

GIV Restrictions From the Binary-Binary Marginal Example

From the restrictions in the table, get upper & lower bounds for each z. Let $P_{11}(z) \equiv \Pr((1,1)|z)$.

$$P_{11}(z) + P_{10}(z) \leq \tau_1 \leq (1 - P_{01}(z)) \quad \Rightarrow$$

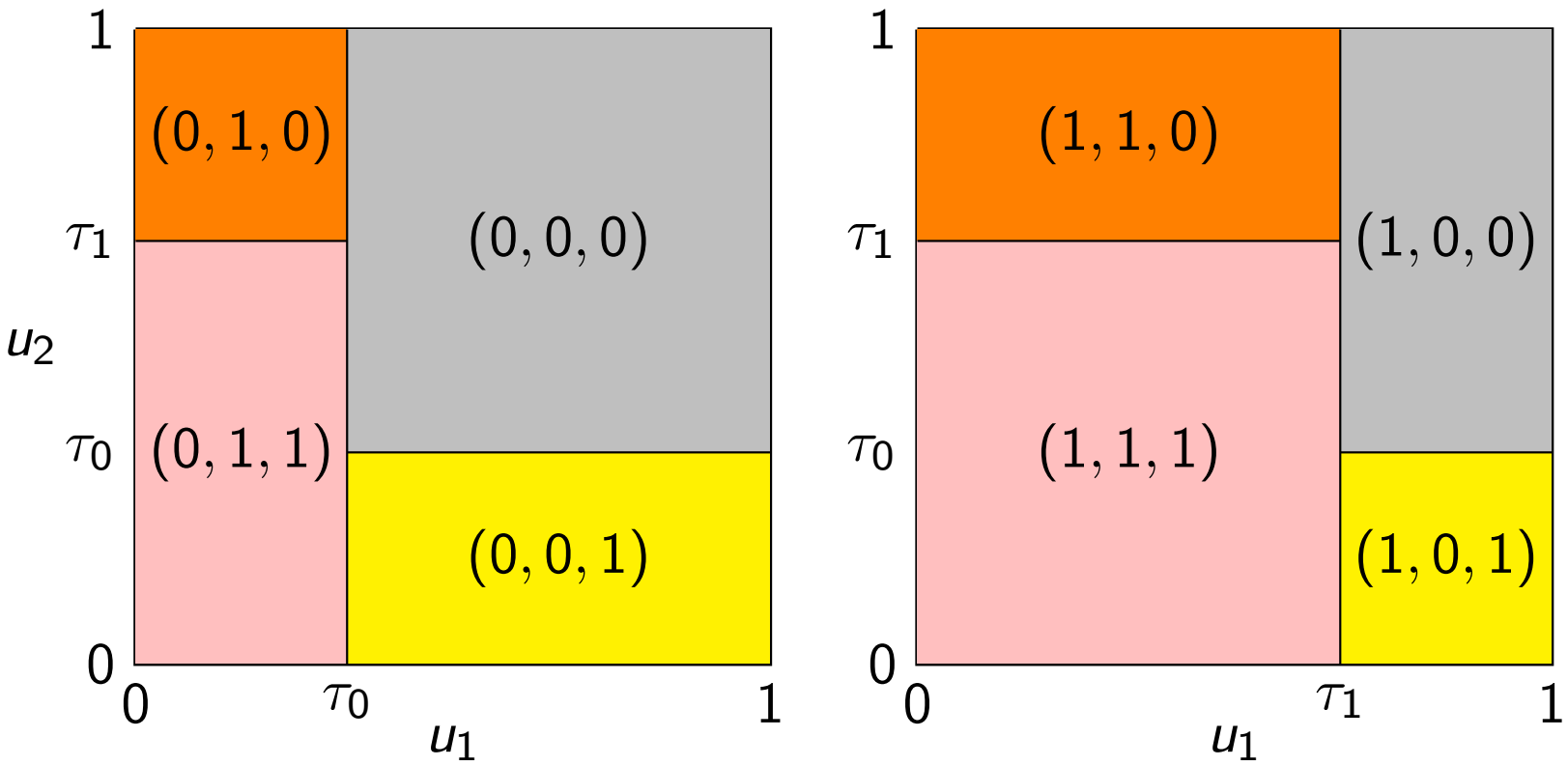
$$\max_z \left[P_{11}(z) + P_{10}(z)\right] \leq \tau_1 \leq \min_z \left[(1 - P_{01}(z))\right].$$

Similar result for τ_0.

But from one transition there is limited information on the structural parameters, and these bounds say nothing about serial correlation.

Elemental Inverse Image Sets, $T = 2$

8 elemental sets $\mathcal{U}(a_i, x_i, \sigma)$, labeled (x_{i1}, a_{i1}, a_{i2})



For example, from the necessary condition for $(0, 0, 0)$

$$\Pr(u_1 > \tau_1, u_2 > \tau_1 | \rho) \geq P_{000}(z) + P_{100}(z) = P_{\cdot 00}$$

Example of Ad-Hoc Upper and Lower Bounds

For example, from the complement (on the unit square) of the $(0, 0, 0)$ set we can get can an upper as well as lower bound:

$$P_{\cdot 00} + P_{11\cdot} \geq \Pr(u_1 > \tau_1, u_2 > \tau_1 | \rho) \geq P_{\cdot 00}$$

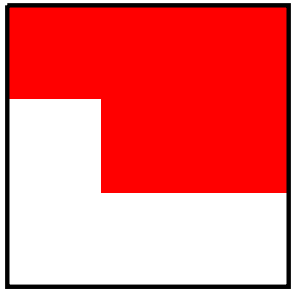
It is easy to generate such upper and lower bounds for the probability on each elemental box on the last slide and only some combinations of (τ_0, τ_1, ρ) will be consistent with all 8 joint probability intervals.

As an extreme case, note that any transition rules out $\rho = 1$ and any weight on $(0, 1, 0)$ rules out ρ close to one.

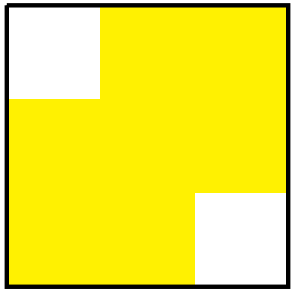
But to get the sharp identified set we have to combine the original 8 elemental necessary conditions with the remaining core determining sets.

Further Core Determining Sets

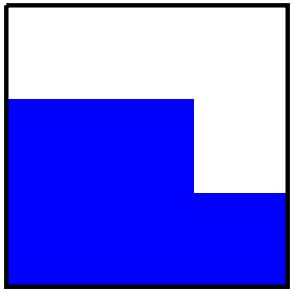
The core determining set includes unions of the "elemental" sets $\mathcal{U}(a_i, x_i, w_i, \sigma)$. The relevant unions are of "partially overlapping" sets.



(a) (0,0,0)+(1,1,0)

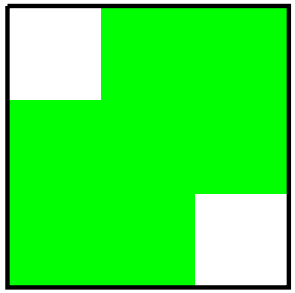


(b) (0,0,0)+(1,1,1)



(c) (1,1,1)+(0,10,1)

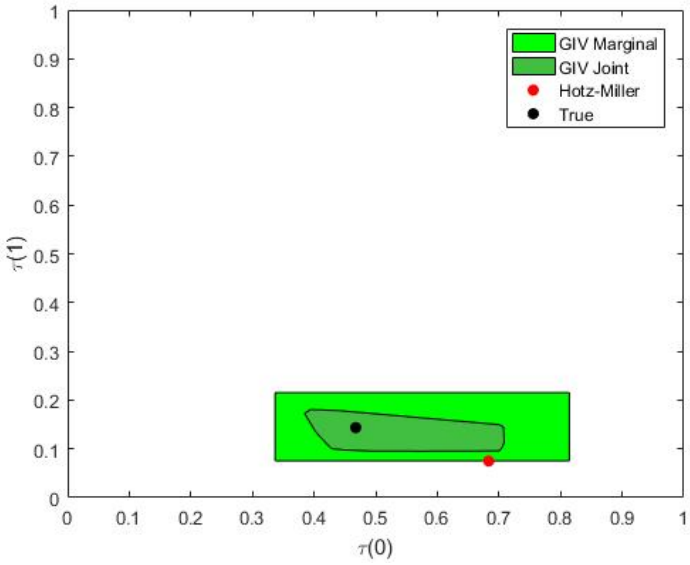
(d) (a)+(b)



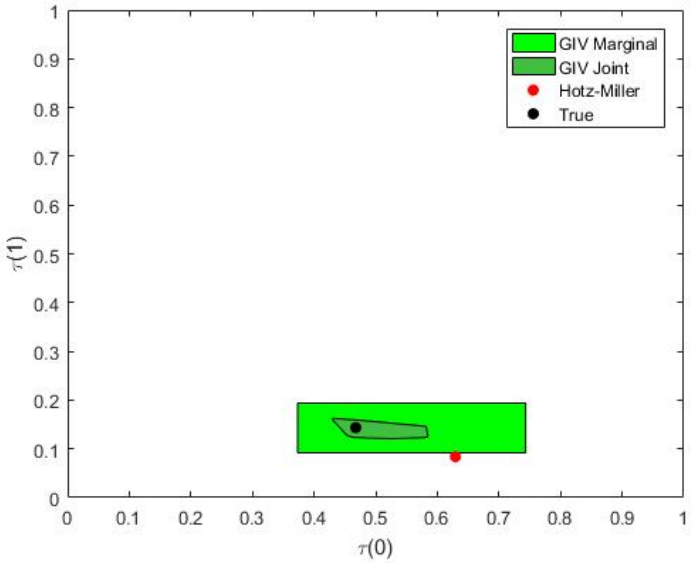
(e) (b)+(c)

Computing the Identified Set of Policy Function Parameters, á la Chesher (2010), etc.

True $\rho = 0.75$, varying IV "strength", Both 1 and 2 periods of data



(a) IV strength = 0.25

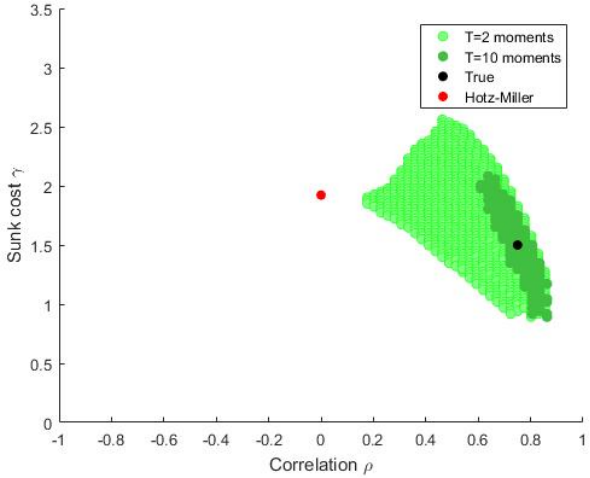


(b) IV strength = 0.56

(Example has $\bar{\pi} = 0.5$, $\gamma = 1.5$. IV "strength" is R^2 from a regression of x_{i1} on z_i. ρ is also a "policy" parameter, see next slide. Policy cutoffs $\tau(a_{it})$ are in the quantiles of ϵ.)

Computing the Identified Set of Structural Parameters

True $\rho = 0.75$, varying IV strength, T=2 or T=10, w_{it} varying or not



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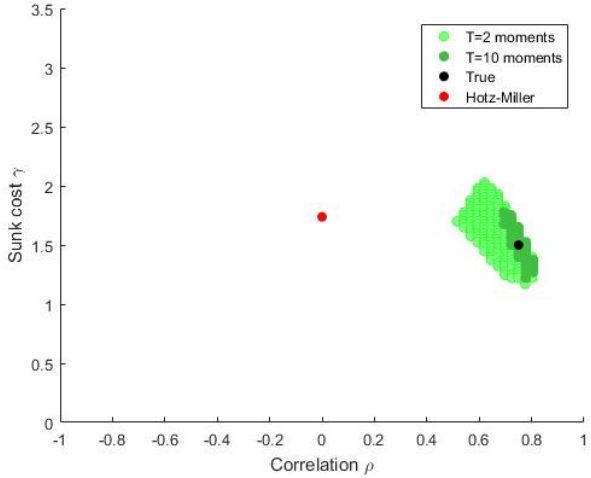
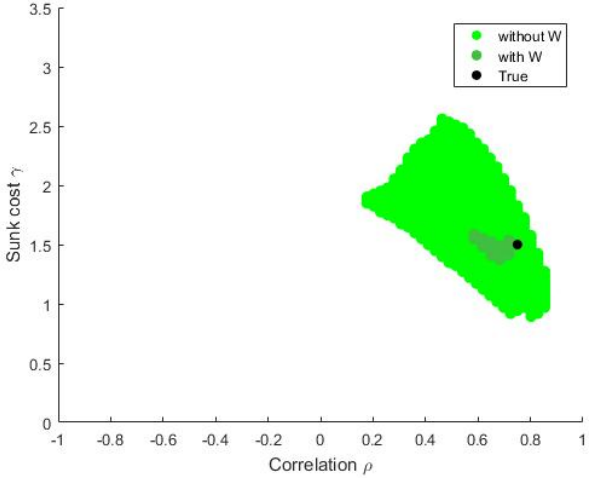


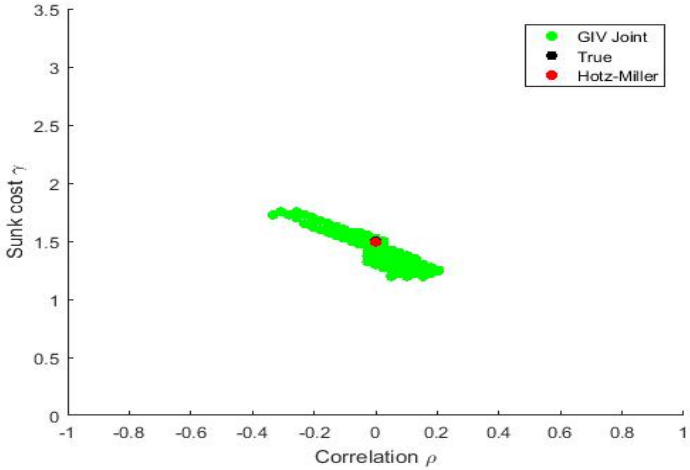
(b) IV strength = 0.56



(c) IV strength = 0.25, varying w_{it}

Computing the Identified Set of Structural Parameters

True $\rho = 0$, varying IV strength, 2 "periods" of data



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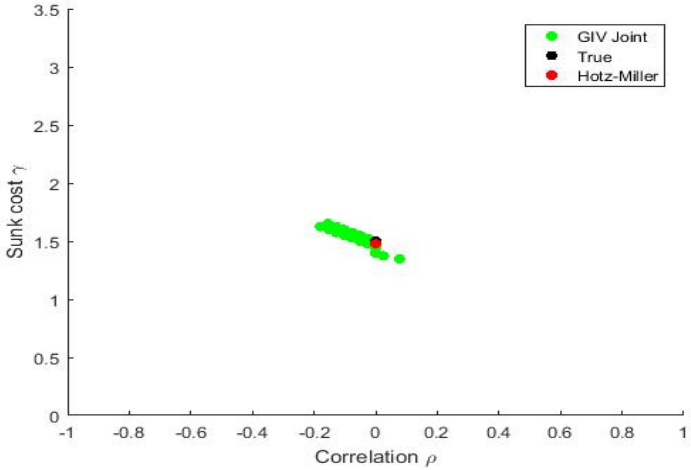
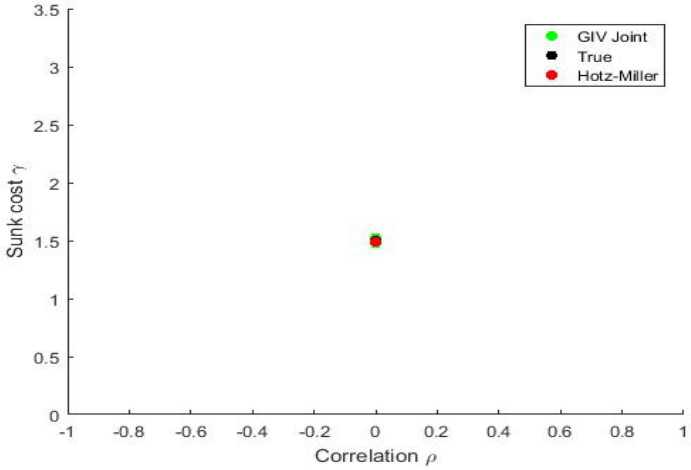


(b) IV strength = 0.56



(c) IV strength = 1.0

Inference

We can apply inference procedures from the moment inequalities literature, simply following on the literature from Chernozhukov, Hong, and Tamer (2007).

With discrete IVs, we work with unconditional moment inequalities (e.g. Andrews and Soares (2010) and Bugni, Canay, and Shi (2017)).

With continuous IVs, we would have conditional moment inequalities (e.g. Andrews and Shi (2013)).

95% Confidence Sets

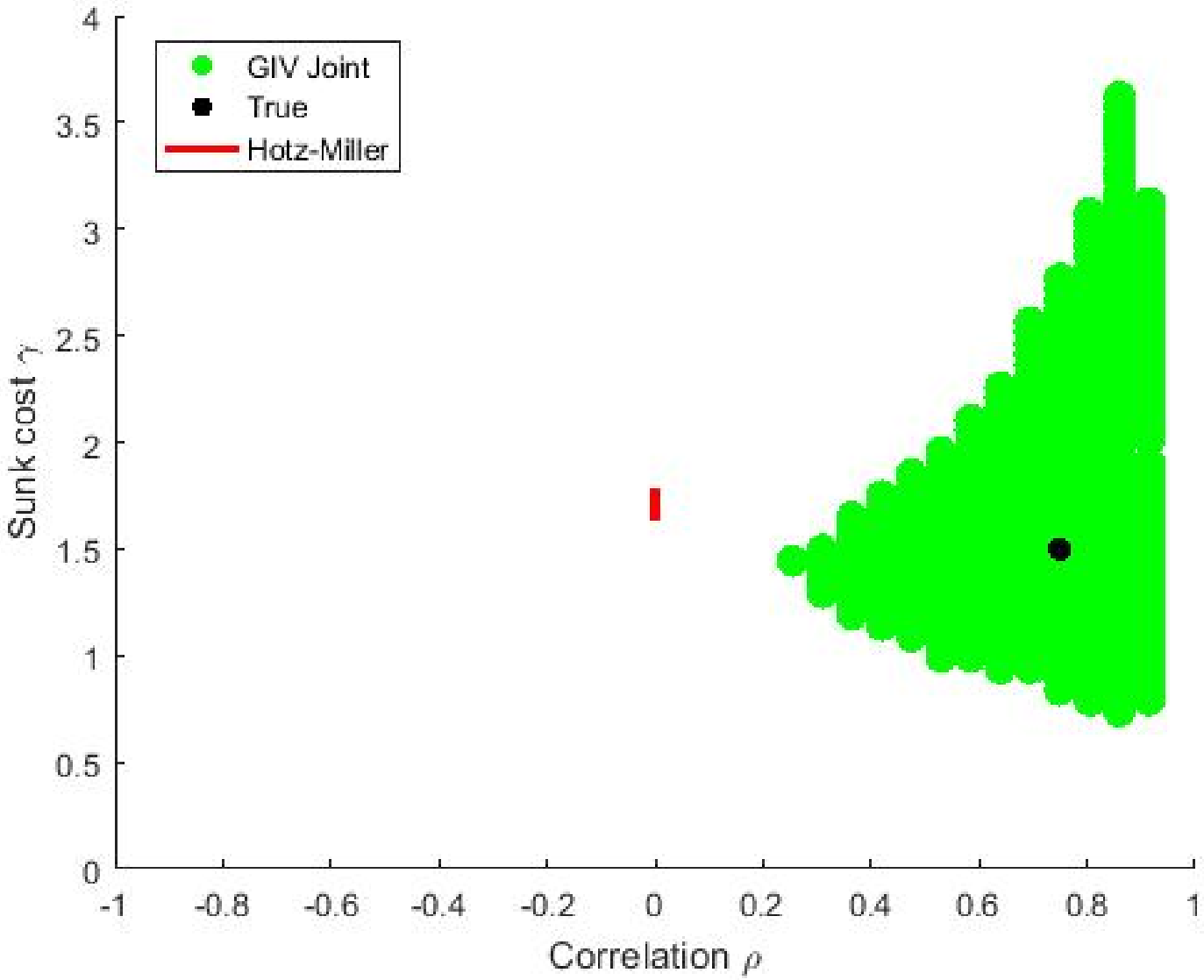


Figure: ρ=0.75, IV strength=0.56

Inference on a Counterfactual 50% Increase in Sunk Costs

Fraction of Markets with an Increase in a_{it} in at Least 1 Period out of the Next 10 Periods

ρ	IV strength	True	GIV Joint	Hotz-Miller
0	0.56	0.44	(0.26,0.58)	(0.41,0.51)
0.50	0.56	0.30	(0.13,0.65)	(0.49,0.70)
0.75	0.56	0.18	(0.05,0.43)	(0.41,0.54)

Dynamic Games

more preliminary

Our approach can be extended to models of dynamic strategic interaction. Specifically, consider dynamic games of complete information and allow for serially correlated unobservables. Let J be the number of players. Then agent j's policy in market i at time t is given by

$$a_{jit} = \sigma_j \left(w_{it}, x_{it}, u_{it} \right) \tag{3}$$

with $w_{it} = \left(w_{1it}, \ldots, w_{Jit} \right)$, etc.

Policies now depend on the unobservables of all players.

With serial correlation, complete information is much easier than private information, but we can add iid private information on top of serially correlated public info, which makes computation (and existence) easier.

Broad Idea of the Games Approach

As in the single agent case, identification comes from combining the GIV restrictions with the Bellman equation that defines the "best response" of each firm to the actions of the other.

Computationally, it is often very difficult to solve for the set of equilibrium strategies. In this case, we can restrict that computation to those strategies that survive the GIV conditions. In a favorable case, this would be a small set.

Additional assumptions may simplify the task, e.g. ensuring the game is symmetric, that strategies are monotonic in some arguments, etc.

Computation in the Oligopoly Case

In some case, it may pay to retain the order of the Hotz-Miller two-step, giving a method that is also close to Ericson and Pakes (1995) but which rules out many possible policy functions via the GIV conditions.

1. Find the set of policies that survive the GIV conditions.
2. For each possible structural parameter vector, see if any of those policies map back into themselves in the dynamic equilibrium. (by holding the rival's policy fixed at a GIV-consistent function and solving the single-firm Bellman equation for the own-firm policy).

Computation in the Oligopoly Case

continued

- Various kinds of monotonicity & symmetry can help a lot.
- We have a worked out computational example with private i.i.d. information plus a discrete serially correlated shock.
- Other good example (not yet computed): continuous oligopoly investment where one might point-identify the policy.

A Hint at An Application

- Begin with the small-town ready-mix concrete example of Collard-Wexler (2014), which is based on public data. The number of concrete firms is shifted by "construction demand," proxied by local construction employment.
- In the spirit of Pakes, Ostrovsky, and Berry (2007), we estimate "variable profit" as a function of the number of firms from (limited) public data, outside of the dynamic model.
- There is a deterministic sunk entry cost & a log-normal per-period fixed cost.
- This leaves us with the four "dynamic parameters:" sunk cost & mean fixed cost, plus the standard deviation & serial correlation of the unobservable.
- We use the Abbring and Campbell (2010) model of the number of firms in dynamic oligopoly, with their "refinement" to unique equilibrium. This makes the problem more like the single-agent case.

Collard-Wexler on Concrete

Collard-Wexler data are for the years 1994 to 2006. He assumes serially correlated errors and effectively estimates the policy-function of the dynamic model, which involve thresholds in the unobservables. He solves the incompleteness problem due to initial conditions by directly modeling the initial unobservables and there is no multiple equilibria problem under the Abbring-Campbell "last in-first out" conditions. He doesn't estimate structural parameters, focusing just on the implied evolution of states over time.

Our VERY Preliminary Work

We extend this to an incomplete model (with an initial conditions problem) and attempt to set-estimate the parameters. For now, we use a very (!) coarse grid and we are using a somewhat ad-hoc set of test sets. There are a number of econometric issues to consider, including whether we should adjust for a large number of moments.

For instruments, we are using long-past interactions of income & population.

Test Sets

There are too many test sets, so we use some simple sets involving, *e.g.*, two-period behavior. We also use some sets that intuitively help to identify (say) sunk cost. For example, the set associated with the event "there is at least one entry in the market over the sample period."

Test Sets

simulation

For some test sets, the probabilities are hard to compute, so use a simulation procedure to obtain test sets. For a given observable event and a given candidate policy function, we

- take many draws for the unobservables
- obtain the equilibrium number of firms over time implied by the candidate policy function
- compute the fraction of markets where the event $\mathcal{E}$ occurs

This gives us the probability on the r.h.s. of the Chesher-Rosen inequality. On the l.h.s. we simply have the probability of the event $\mathcal{E}$, conditional on values of the instruments.

Very Preliminary Results

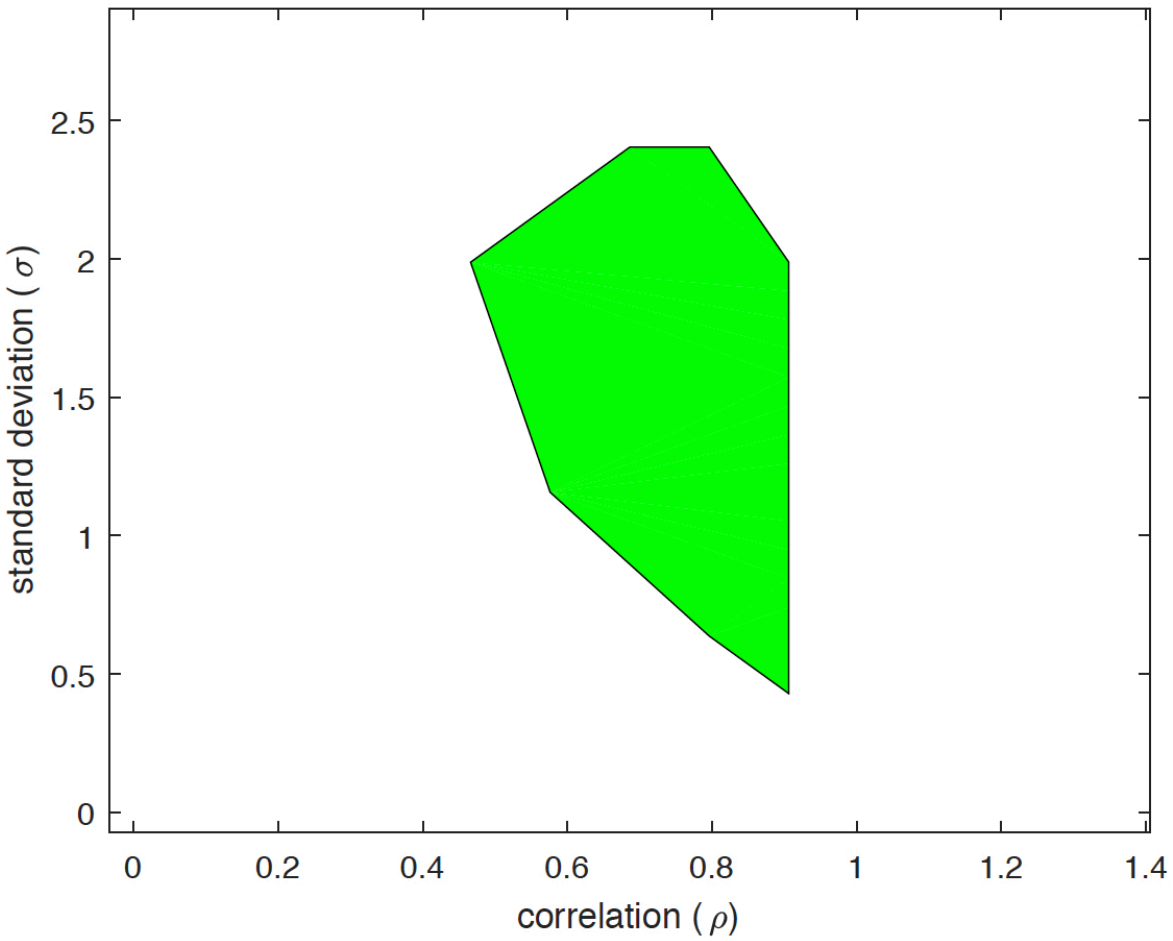


Figure: Concrete Data: Prelim Projection of the Estimated Set for (ρ, σ)

Very Preliminary Results

continued

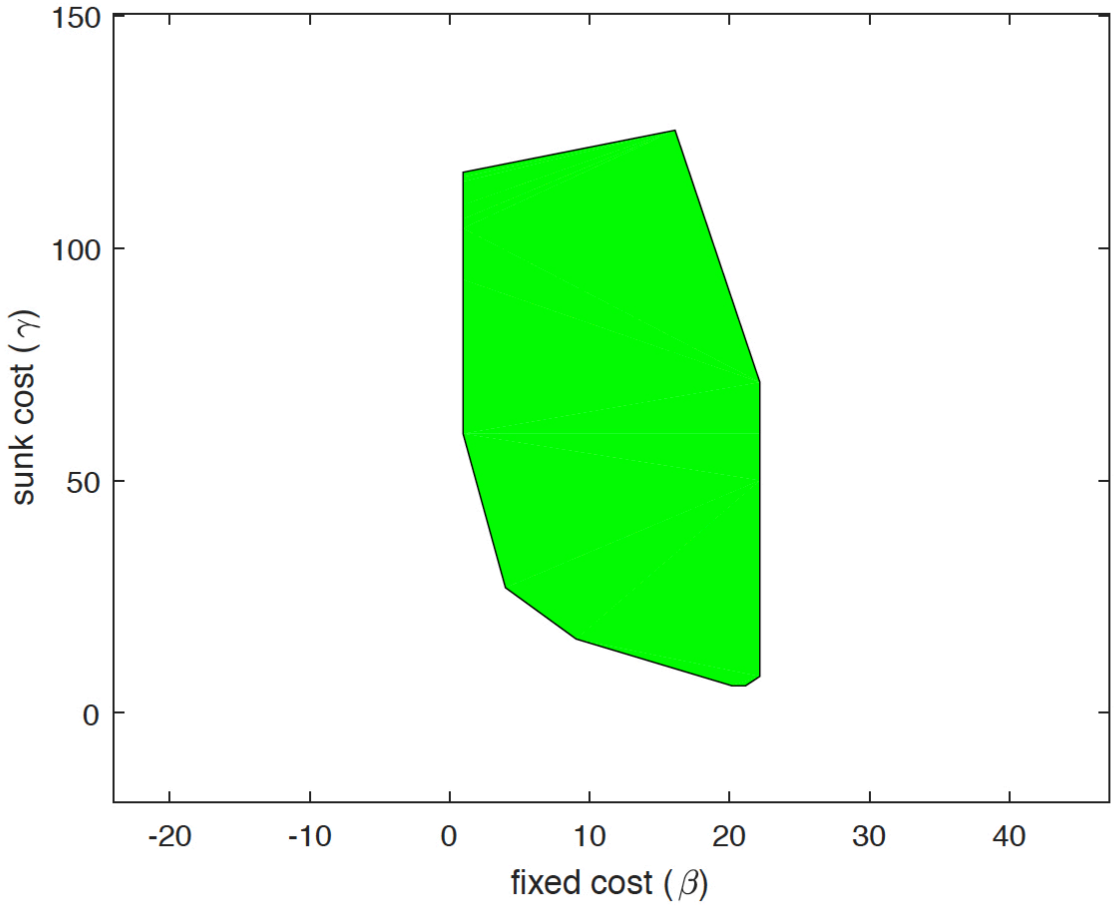


Figure: Concrete Data: Prelim Projection of the Estimated Set for (β, γ)

Initial Conditions in the Single-Agent Case

In our single-agent case, as in Collard-Wexler, we could complete model with an *ad-hoc* equation (or process) that determines the initial x_{i1}, for example,

$$x_{i1} = f(z_i, u_{i0})$$

This would give us equalities in condition (1) and then in simple discrete cases the GIV method is equivalent to MLE. This gives an "IV" interpretation to MLE applied to the complete model. Note: MLE might or might not be point identified, past exogenous shifters can still be important.

An Alternative: Dynamic Panel Restrictions

Note that our model of serial correlation

$$u_{it} = \rho u_{t-1} + \sigma \nu_{it}$$

suggests an additional "dynamic panel data" style restriction, that the "innovation" ν_{it} is independent (or mean independent) of past *endogenous* states $(x_{it-1}, x_{it-2}, \ldots)$.

This "dynamic panel" restriction is not really in the spirit of the current paper, but it suggests an approach that may be very helpful when instruments for current market structure are not available or not very powerful.

A Last Practical Concern

By a classic curse of dimensionality in dynamic games, the number of "natural" parameters in the policy function grows rapidly in the number of states and players. However, this problem also occurs in the "exogenous states" i.i.d. literature. In both cases, we need some lower dimensional approximation to the policy functions.

The existing empirical literature often uses an (internally inconsistent) "double parametric" specification, with (for example) a "nonparametric" entry equation estimated in the first stage by a highly parametric probit equation. Our approach suggests, at least, estimating that probit via GIV methods rather than MLE treating states as econometrically exogenous.

Conclusion

- Market structure in dynamic IO models should not be assumed to be exogenous.
- General IV models are one natural way to handle the econometric endogeneity of states, while preserving much of the intuition of existingt ggggwo-step methods.
- Intuition of IV: past exogenous shifters (and regulation, etc) are correlated with today's state.
- May have set-identified policies and/or structural parameters. The data, the IV restriction and Bellman's equation together restrict the identified set.

Needed: more applications!

References I

Abbring, J. H., and J. R. Campbell (2010): "Last-In First-Out Oligopoly Dynamics," *Econometrica*, 78(5), 1491–1527.

Andrews, D. W., and X. Shi (2013): "Inference based on conditional moment inequalities," *Econometrica*, 81(2), 609–666.

Andrews, D. W., and G. Soares (2010): "Inference for Parameters Defined by Moment Inequalities Using Generalized Moment Selection," *Econometrica*, 78(1), 119–157.

Arcidiacono, P., and R. Miller (2011): "Conditional Choice Probability Estimation of Dynamic Discrete Choice Models with Unobserved Heterogeneity," *Econometrica*, 7(6), 1823–1868.

References II

Bajari, P., C. L. Benkard, and J. Levin (2007): "Estimating Dynamic Models of Imperfect Competition," *Econometrica*, 75(5), 1331–1370.

Beresteanu, A., F. Molinari, and I. Molchanov (2011): "Sharp Identification Regions in Models with Convex Moment Predictions," *Econometrica*, 79(6), 1785–1821.

Berry, S., and E. Tamer (2007): "Identification in Models of Oligopoly Entry," in *Advances in Economics and Econometrics: Theory and Applications, Ninth World Congress*, ed. by W. N. R. Blundell, and T. Persson, vol. 2. Cambridge University Press.

Bugni, F. A., I. A. Canay, and X. Shi (2017): "Inference for subvectors and other functions of partially identified parameters in moment inequality models," *Quantitative Economics*, 8(1), 1–38.

References III

Chernozhukov, V., and C. Hansen (2005): "An IV Model of Quantile Treatment Effects," *Econometrica*, 73(1), 245–261.

Chernozhukov, V., H. Hong, and E. Tamer (2007): "Estimation and Confidence Regions for Parameter Sets in Econometric Models1," *Econometrica*, 75(5), 1243–1284.

Chesher, A. (2010): "Instrumental Variables Models for Discrete Outcomes," *Econometrica*, 78, 575–601.

Chesher, A., and A. Rosen (2017): "Generalized Instrumental Variable Models," *Econometricas*, 83(3), 959–989.

Ciliberto, F., and E. Tamer (2009): "Market Structure and Multiple Equilibria in Airline Markets," *Econometrica*, 77(6), 17911828.

References IV

Collard-Wexler, A. (2014): "Mergers and Sunk Costs: An Application to the Ready-Mix Concrete Industry," *American Economic Journal: Microeconomics*, 6(4), 407–447.

Ericson, R., and A. Pakes (1995): "Markov Perfect Industry Dynamics: A Framework for Empirical Work," *Review of Economic Studies*, 62, 53–82.

Galichon, A., and M. Henry (2011): "Set identification in models with multiple equilibria," *Review of Economic Studies*, 78(4), 1264–1298.

Hotz, J., and R. A. Miller (1993): "Conditional Choice Probabilites and the Estimation of Dynamic Models," *Review of Economic Studies*, 60, 497–529.

Kalouptsidi, M., P. Scott, and E. Souza-Rodrigues (2018): "Linear IV Regression Estimators for Single-Agent Dynamic Discrete Choice Models," Discussion paper, Harvard.

References V

Kasahara, H., and K. Shimotsu (2009): "Nonparametric Identification of Finite Mixture Models of Dynamic Discrete Choices," *Econometrica*, 77(1), pp. 135–175.

Keane, M. P., and K. I. Wolpin (1997): "The Career Decisions of Young Men," *Journal of Political Economy*, 105(3), 473–522.

Magnac, T., and D. Thesmar (2002): "Identifying Dynamic Discrete Decision Processes," *Econometrica*, 70(2), 801–816.

Manski, C. F. (2003): *Partial Identification of Probability Distributions*. Springer, New York.

Manski, C. F., and E. Tamer (2002): "Inference on Regressions with Interval Data on a Regressor or Outcome," *Econometrica*, 70(2), 519–546.

References VI

Pakes, A., and R. Ericson (1996): "Empirical Implications of Alternative Models of Firm Dynamics," *Journal of Economic Theory*.

Pakes, A., M. Ostrovsky, and S. Berry (2007): "Simple Estimators for the Parameters of Dynamic Games, with Entry/Exit Examples," *RAND Journal of Economics*, 38(2), 373–399.

Pesendorfer, M., and P. Schmidt-Dengler (2008): "Asymptotic Least Squares Estimators for Dynamic Games," *Review of Economic Studies*, 75(3), 901–928.

Rust, J. (1987): "Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher," *Econometrica*, 55(5), 999–1033.

References VII

Tamer, E. (2003): "Incomplete Simultaneous Discrete Response Model with Multiple Equilibria," *The Review of Economic Studies*, 70(1), pp. 147–165.